MILITARY SPECIFICATION

CHISEL, PERIODONTAL (Double-Ended)

This specification is approved for use within DLA, Defense Personnel Support Center (DM), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a double-ended chisel for use in periodontal surgical and scaling procedures.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

GG-I-526 - Instruments, Dental and Surgical; General Specifications.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Military

MIL-P-116 - Preservation, Methods of.
MIL-L-10547 - Liners, Case, and Sheet; Overwrap; WaterVaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

Drawings

Defense Personnel Support Center (DPSC)

24490 Handles, Dental, Periodontal Instruments
24519 Chisel, 13K and 13KL

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society for Testing and Materials (ASTM)

ASTM A 473 - Standard Specification for Stainless and Heat-Resisting Steel Forgings.
ASTM B 135 - Standard Specification for Seamless Brass Tube.
ASTM E 18 - Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
ASTM E 92 - Standard Test Method for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The chisel shall be in accordance with applicable requirements of GG-I-526 with the options and additional requirements specified herein.

3.2 Material.

3.2.1 Operative ends. The operative ends shall be class 4, type 440A, 440B, or 440C stainless steel.

3.2.2 Handle. Handle A and D shall be plated brass. Brass shall conform to class 8, copper alloy number 330, as specified in ASTM B 135 for chemical composition only. The brass handles shall be chromium over nickel plated. Handle B shall be fabricated of martensitic stainless steel, type 416 or 410, as specified in GG-I-526, or ferritic stainless steel conforming to 430F stainless steel in accordance with ASTM A 473 for composition only. Handle C shall be fabricated of austenitic stainless steel, type 303, as specified in GG-I-526. The corrosion tests, as specified in 3.8, shall apply as applicable.

3.3 Hardness. After proper stress relieving, the Rockwell hardness of the operative ends shall be C52 to C56 for type 440A, C54 to C58 for type 440B, and C56 to C60 for type 440C. Handles and shanks shall be of proper temper and shall have a hardness suitable for the purpose intended. Hardness shall be determined as specified in 4.4.1.

3.4 Style, design and dimensions. Shall be a double-ended Kirkland 13K and 13KL pull chisel suitable for use in periodontal osseous surgery and as a heavy scaler of teeth and exposed root surfaces. The chisel shall be furnished with the handle and shank design of either handle A, B, C or D conforming to Defense Personnel Support Center (DPSC) Drawing 24490. The style, design and dimensions of the chisel shall conform to DPSC Drawing 24519.

3.5 Construction. The chisel shall be of three piece construction consisting of a handle and an operative end at each end. Handle shall be of solid or hollow construction as specified on the drawing for the respective handle. Knurled or serrated portions of handles shall be as specified on the drawing for the respective handle.

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The operative ends shall be permanently secured to the handle. Junction of operative end to handle shall be smoothly finished and free of crevices. There shall be no movement, loosening, or separation between the operative end and the handle when the operative end is subjected to an 85 pound (minimum) pull out force, when tested as specified in 4.4.2.

3.5.1 Drawings. The drawings forming a part of this specification are not production drawings. Drawings establish dimensional limitations and control design and configuration. The supplier shall be responsible for preparing his own shop drawings. The supplier shall select dimensions and tolerances within the limitations prescribed on the drawings to provide a product with proper design, configuration, and function. No deviation from the dimensional limitations prescribed on the drawings is permissible without prior approval of the contracting officer.

3.6 Finish. The operative ends, including shanks, shall have a smooth mirror or smooth bright finish, except for sharpened edges which shall have a finely honed finish. The operative ends shall have a sharp cutting edge and shall be free of defects, such as coarse grind marks, cracks, gouges, burrs, feathers, or other imperfections which affect performance, when viewed as specified in 4.4.3. Finger grip portions of handles shall be serrated or knurled. Handles shall have a mirror or bright finish, except for the serrated or knurled portions. Finger grip portions of handles shall be free from barbs and shall provide adequate finger grip during use.

3.7 Corrosion resistance. The copper sulfate test requirements of GG-I-526 shall apply to the operative ends, martensitic stainless steel handles, and ferritic stainless steel handles. The boil test requirements of GG-I-526 shall apply to the entire chisel (operative ends and all handles). The salt spray test specified for brass composition in GG-I-526 shall not apply.

3.8 Identification marking. In addition to the requirements of GG-I-526, each chisel shall be permanently and legibly marked with the respective Kirkland instrument numbers, as applicable. All marking shall be on the smooth portion of the handle. Each chisel shall be marked with the name or registered trademark of the manufacturer (and contractor, if other than the manufacturer). The word "stainless", when applicable, shall also be marked on instruments of foreign manufacture. As an alternate to the word "stainless", the use of a registered trade name identifying the steel as stainless shall be acceptable, provided prior approval is received from the cognizant contracting officer.

3.9 Workmanship. In addition to the requirements of GG-I-526, the operative ends shall be free of deep pits, discoloration, and rounded edges.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the raw material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables I, II, III (see 4.2.1, 4.2.2, 4.3), and 4.3.1.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in table I. The unit of product for examination shall be one chisel.

TABLE I. Sampling for examination.

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	4.0

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4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in table II. The unit of product for test purposes shall be one instrument. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE II. Sampling for tests.

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL
Entire instrument	Composition	3.2.1, 3.2.2, & GG-I-526	GG-I-526	S-1*	**
Operative end	Hardness	3.3 and GG-I-526	4.4.1	S-2	1.5
Operative end	Dimensions	3.4	SMI ***	S-2	2.5
Handle, shank, Overall instrument	Dimensions	3.4	SMI ***	S-4	4.0
Operative end and handle	Pull test	3.5	4.4.2	S-2	1.0
Cutting edges	Magnification inspection	3.6	4.4.3	I	**
Operative end and handle (martensitic and ferritic)	Copper sulfate test	3.7	GG-I-526	S-2	1.0
Entire instrument	Boil test	3.7	GG-I-526	S-2	1.0

* A sufficient amount of each lot of metal used in the construction of the chisels may be selected for composition analysis in lieu of samples chisels, provided lot identity has been maintained and can be demonstrated to the Government.

** Acceptance number shall be zero for all sample sizes.

*** Measurements shall be taken utilizing Standard Measuring Instruments (SMI).

4.3 Examination. The chisels shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with the following classification of defects.

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Examination shall include, but shall not be limited to, the defects listed in table III.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Style and design not as specified.
102	Instrument not constructed as specified
103	Operative end finish not as specified.
Minor	
201	Instrument not permanently and legibly marked with name or trademark of the contractor.
202	Instrument identification not present, properly located, or not complete.
203	Curette not free from dirt, oil, compound, or foreign matter.
204	Handle serrations or knurling missing, not uniform, or not complete.
205	Junction of handle not smoothly finished and free of crevices.

4.3.1 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with the requirements of this specification (see 4.2.2.).

4.4.1 Hardness. Hardness shall be determined in accordance with ASTM Designation E 18 (Rockwell) or E 92 (Vickers or Diamond Pyramid).

4.4.2 Pull test. The handle and shank of the operative end shall be securely held in separate chucks or other holding devices. A slowly increasing pull out force, increasing from zero to 85 pounds (minimum), shall be applied to the shank of the operative end. The force shall be maintained at the high value for a minimum of 1 second. The instrument shall then be examined for movement, looseness, or separation of the operative end with respect to the handle. The test shall be repeated for each operative end of the double-ended instrument.

4.4.3 Magnification inspection. The operative ends (blade portion) shall be inspected under four power (4X) magnification.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1. Cleaning and drying. Each chisel shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. Each chisel shall be wrapped in neutral tissue or other suitable non-corrosive barrier material, and packaged in accordance with MIL-P-116, method IC-2. As an alternate, the unit package shall be a rigid plastic tube having a metal screw cap or a plastic friction plug type closure. The chisel shall be cushioned within the tube in such a manner to prevent damage to the operative ends.

5.1.1.4 Intermediate package. Twelve chisels shall be packaged in a box conforming to PPP-B-566, PPP-B-636, type CF, class domestic, or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for chisels shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B and C, as specified (see 6.1).

5.2.1 Level A. Twenty-four intermediate packages (288 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2 or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Twenty-four intermediate packages (288 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1 or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packaging for chisels shall conform to the MIL-STD-794 requirements for this level.

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5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length, and shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Chisels packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

5.2.5.2 Level C. Chisels packed as specified in 5.2.3 shall be unitized as specified in 5.2.5.

5.3 Marking.

5.3.1 Level A, B, and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.3.2.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following item listed in the Federal Supply Catalog:

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National Stock Number	Item Identification
6520-01-093-2280	CHISEL AND SCALER, PERIODONTAL, Double-ended, Kirkland, No. 13K and 13 KL.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

Project No. 6520-1918

Location: NCR

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-42003 (DM)	CHISEL, PERIODONTAL (Double-Ended)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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